

INNOVATION PROPOSAL

SkillMatch AI

Your Personal AI Career Mentor

Bridging the gap between talent and opportunity through AI-powered, personalized career intelligence.

Submission: IDEA CONTEST

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1. Executive Summary

SkillMatch AI is an intelligent career-mentorship platform that uses artificial intelligence to guide students and early-career professionals from “what should I learn?” to “where should I work?”. It combines skill-gap analysis, personalized learning roadmaps, real-time job-market intelligence, and AI-driven mock interviews into a single mobile and web application — replacing fragmented advice from social media, generic counsellors, and unstructured YouTube tutorials with a data-driven, one-to-one AI mentor available 24/7.

Bangladesh graduates over 800,000 students each year, yet nearly one in three remains jobless for up to two years after graduation. The core reason is not a shortage of jobs — it is a shortage of the right skills, the right guidance, and the right feedback loop. SkillMatch AI attacks this problem directly: it analyses each user’s skills, compares them against live market demand, generates a personalized roadmap, tracks progress, and prepares the user for real interviews.

The platform is designed to be mobile-first, affordable, and inclusive — usable by students in Tier-1 universities such as RUET, but equally usable by rural learners with mid-range smartphones. Our target for Year 1 is 50,000 active users across Bangladesh, followed by regional expansion into South Asia in Year 2.



Figure 1. AI-powered platforms are transforming how students discover careers and build employable skills. (Source: NACE)

2. Problem Statement

Despite the explosion of online learning resources, students and job-seekers face a paradox of choice combined with a lack of direction. The result is high unemployment among educated youth, high under-employment among fresh graduates, and a widening skill-market mismatch across the country.

2.1 Key Pain Points

- **Fragmented guidance:** Advice is scattered across LinkedIn posts, YouTube videos, Facebook groups, and coaching centres — none of it personalized to the individual student.
- **Skill-market mismatch:** Curricula lag behind industry demand in areas such as cloud computing, data engineering, prompt engineering, and modern DevOps.
- **Costly and inaccessible mentorship:** Quality 1-to-1 career mentors cost more than most students can pay, and are almost non-existent outside major cities.
- **Poor interview readiness:** Students memorize theory but freeze in real interviews because they have never practiced under realistic conditions with objective feedback.

- **Inefficient recruitment:** Recruiters spend weeks filtering unstructured CVs; qualified candidates get missed because their skills are not surfaced in a searchable format.



Figure 2. Fresh graduates face a challenging job market — a structural opportunity for SkillMatch AI. (Source: The Business Standard)

3. Proposed Solution

SkillMatch AI is a personal AI career mentor delivered as a mobile app (with a companion web dashboard) that continuously learns about the user and provides tailored, actionable career guidance. The platform performs three continuous functions:

- **Assess.** Build a live skill profile from CV parsing, quizzes, coding tests, and self-reported goals.
- **Guide.** Generate a step-by-step learning roadmap mapped to real job openings in the target market.
- **Connect.** Match users with relevant internships and jobs, then prepare them for interviews through AI-driven mock sessions.

Every interaction feeds a feedback loop that continuously refines the user's roadmap — turning the platform into a genuine mentor rather than a static content library.

4. Objectives

- Deliver personalization at scale — a unique roadmap for every user, generated in seconds.
- Close the skill gap between what universities teach and what employers hire for.
- Improve employability of fresh graduates and reduce the time-to-first-job.
- Enable data-driven institutions — universities and training centres — to see skill trends in real time.
- Build an inclusive, mobile-first platform accessible to rural and under-served students.

5. Key Features

- **AI Skill-Gap Analyzer.** Parses CV, projects, and quiz results; benchmarks against live job postings; produces a numeric skill-fit score for chosen career tracks.
- **Personalized Learning Roadmap.** Recommends specific courses, projects, and micro-certifications with estimated completion time — ordered by return-on-effort.
- **Real-Time Job & Internship Matching.** Scrapes and normalizes job feeds from LinkedIn, Bdjobs, Skill.jobs, and company career pages; ranks matches by skill compatibility.
- **AI Mock Interview Simulator.** Voice-based interviews using LLMs; scores answers on content, structure, clarity, and confidence; produces improvement notes.
- **24/7 Career Chat Mentor.** Conversational assistant that answers career questions in Bangla or English with hyper-local context (universities, companies, salary ranges).

- **Progress & Analytics Dashboard.** Visualizes skill growth, streaks, weekly learning hours, and readiness for target roles.
- **Institutional & Recruiter Portal.** Universities see aggregated skill trends of their students; recruiters post jobs and access ranked, verified candidate pools.

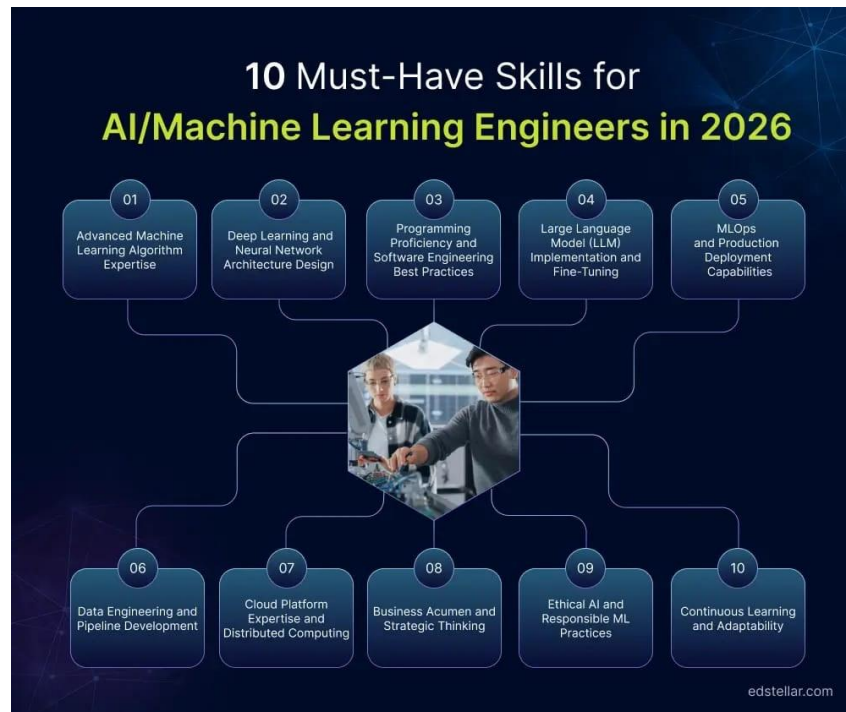


Figure 3. SkillMatch AI maps user competencies to the most in-demand AI, cloud, and software-engineering skills. (Source: Edstellar)

6. System Workflow

The user journey through SkillMatch AI follows a seven-stage cycle:

- **Step 1 — Onboarding & Profiling.** User signs up, uploads CV, and answers a short career-interest survey.
- **Step 2 — AI Skill-Gap Analysis.** The engine extracts skills, benchmarks them against target roles, and computes a fit score.
- **Step 3 — Roadmap Generation.** LLM + recommendation models generate a weekly plan of courses, projects, and milestones.
- **Step 4 — Skill Development Tracking.** Users log activities; badges and streaks reinforce consistency.
- **Step 5 — Job & Internship Matching.** Once fit-score crosses a threshold, matching jobs are surfaced automatically.
- **Step 6 — Interview Preparation.** AI mock interviews plus curated question banks tailored to matched roles.
- **Step 7 — Feedback Loop.** Outcomes (offers, rejections, interviewer feedback) retrain the model — the mentor gets smarter with every user.

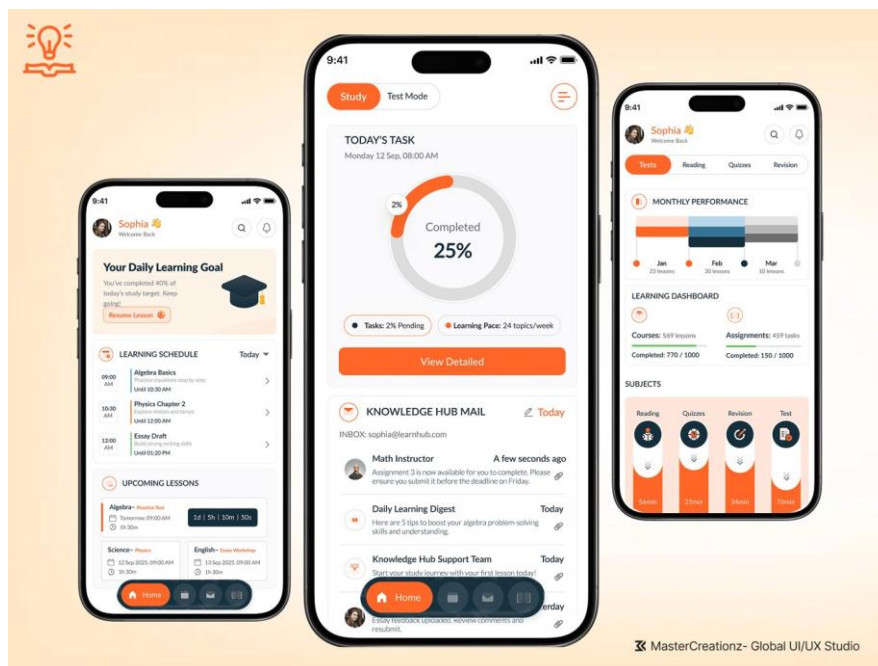


Figure 4. Concept of the SkillMatch AI dashboard — progress, goals, and analytics in one glance. (Source: Master Creationz / Dribbble)

7. Technical Architecture

7.1 Technology Stack

Layer	Technology / Framework
AI / ML core	LLM-based conversational mentor (GPT-class / Llama); Random Forest & Gradient Boosting for skill-gap prediction; collaborative filtering for job matching
Backend	Node.js (Express) and Python (FastAPI) microservices
Mobile frontend	Flutter (single codebase for Android + iOS)
Web dashboard	React.js + Tailwind CSS
Databases	PostgreSQL (structured); Pinecone / FAISS (vector search on skills & jobs)
Cloud & DevOps	AWS / Google Cloud / Azure; Docker; CI/CD via GitHub Actions
Security	OAuth 2.0, JWT, end-to-end encryption of user data

7.2 Data Sources

- Job aggregators: LinkedIn, Bdjobs, Skill.jobs, and public company career pages.
- Learning content: Coursera, edX, YouTube EDU, 10 Minute School, and open-source curricula.
- User-generated: CVs, project links, quiz responses, mock-interview transcripts.

AI Tech Stack: Components

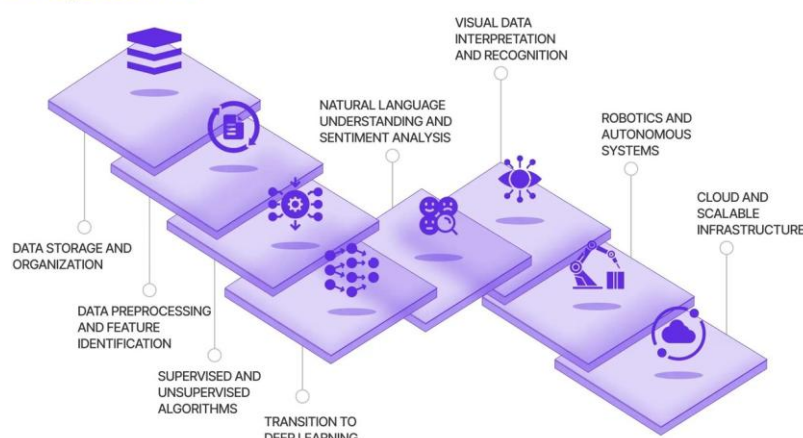


Figure 5. Reference architecture concept — a modern AI application stack combining LLMs, vector search, and cloud services.
(Source: Markovate)

8. Innovation & Uniqueness

Existing Solutions	Limitation	SkillMatch AI Advantage
LinkedIn Learning	Passive content library, no roadmap	Personalized, adaptive roadmap tied to live jobs
Coursera / edX	Course-centric, no job feedback loop	Learning is anchored to real matched opportunities
Career counsellors	Expensive, subjective, city-limited	AI mentor available 24/7, affordable, nationwide
Job portals (Bdjobs, etc.)	List jobs but do not upskill users	Upskills users until they qualify for target roles

SkillMatch AI is the first solution in the Bangladeshi context to close the entire loop — assess, learn, apply, interview, retrain — inside a single application, powered by locally-tuned AI models that understand the country’s universities, employers, and salary norms.

9. Target Users

- University students (all disciplines) preparing for internships and first jobs.
- Fresh graduates within 0–2 years of graduation.
- Early-career professionals (0–5 years) planning a career switch or upskilling.
- Educational institutions seeking data on their students’ skill profile.
- Recruiters and HR teams looking for pre-screened, skill-verified candidates.

10. Social Impact

- **Lower graduate unemployment.** Faster, better matches between talent and opportunity across Bangladesh.
- **Modernized curricula.** Universities receive live skill-gap analytics they can use to update syllabi.
- **Inclusion of rural talent.** A mobile-first, low-bandwidth design brings quality mentorship to students outside metropolitan areas.

- **Empowered women in tech.** Private, on-demand mentorship removes many of the social barriers women face in traditional coaching environments.
- **Contribution to SDGs.** Directly aligned with UN SDG 4 (Quality Education), SDG 8 (Decent Work), and SDG 9 (Industry, Innovation & Infrastructure).

11. Business Model

Revenue Stream	Description	Estimated Share
Freemium subscription	Free core features; premium tier for advanced mock interviews and analytics	45%
Institutional licensing	Annual license for universities and training centres	30%
Recruiter partnerships	Access to verified, ranked candidate pools	15%
Sponsored learning content	EdTech partners sponsor curated learning paths	10%

The core mentoring loop remains free forever, ensuring that no student is excluded on the basis of ability to pay. Revenue is generated primarily from institutions and premium features that add convenience, not from gate-keeping essential guidance.

12. Implementation Plan

Phase	Timeline	Key Deliverables
Phase 1 — Research & Design	Month 1–2	User research, UX prototypes, dataset preparation
Phase 2 — MVP Development	Month 3–5	Core skill analyzer, roadmap engine, mobile MVP
Phase 3 — Pilot Testing	Month 6–7	Closed beta with 2 universities (including RUET)
Phase 4 — Public Launch	Month 8–9	Play Store & App Store release; marketing kickoff
Phase 5 — Scale & Expand	Month 10+	Institutional partnerships, regional expansion, AR/VR interviews

13. Feature Spotlight: AI Mock Interview

The AI Mock Interview module is a signature feature. Users select a target role — Software Engineer, Data Analyst, ML Engineer, Product Manager, and so on — and enter a realistic voice-based interview with an AI recruiter. Responses are scored on five axes: technical accuracy, communication clarity, structure, confidence, and vocabulary. A written improvement report is generated within seconds of the session ending, along with recommended micro-lessons for weak areas.



Figure 6. AI Mock Interview — realistic, on-demand practice with instant, structured feedback. (Source: FavTutor)

14. Challenges & Mitigation

Challenge	Mitigation Strategy
Data privacy of user profiles	End-to-end encryption; explicit consent; local storage of sensitive fields
AI accuracy & bias	Continuous retraining with human-in-the-loop; fairness audits
Low digital literacy of some users	Chat-first, Bangla-supported UI; guided onboarding tutorials
Monetization vs. accessibility	Core mentoring loop remains free; premium only for optional features
Cold-start on job data	Bootstrapped via public feeds and partnerships with Bangladeshi employers

15. Market Opportunity

Bangladesh has one of the youngest populations in Asia, with more than 4.4 million students enrolled in tertiary education and over 800,000 graduates entering the workforce each year. Combined with a rapidly digitizing economy and rising demand for tech-skilled talent, this creates a large and fast-growing addressable market for SkillMatch AI.

Market Segment	Size (Bangladesh)	Relevance to SkillMatch AI
University students	~4.4 million	Primary user base
Annual graduates	~800,000 / year	Recurring inflow of new users
Registered universities	~160+	Institutional licensing prospects
Registered recruiters (Bdjobs)	~30,000+ companies	Recruiter portal customers
Regional expansion (South Asia)	~40 million students	Year-2 growth market

Even at a conservative 3% penetration of Bangladeshi graduates in Year 1, SkillMatch AI can reach 25,000+ active users; scaling to 5% in Year 2 with institutional partnerships puts the platform on a clear path to nation-wide reach and financial self-sufficiency.

16. Expected Outcomes & Success Metrics

KPI	Year 1 Target	Year 2 Target
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Active users	50,000	250,000
University partnerships	5	25
Recruiter partners	30	150
Average skill-fit score improvement	+25 points	+40 points
Reduction in time-to-first-job (self-reported)	-20%	-35%
User retention (30-day)	55%	70%
Mock interviews completed	200,000	1,000,000

Each KPI is measurable directly inside the platform, giving both the team and institutional partners a transparent view of impact — from individual user growth to macro-level employability outcomes.

17. Team & Execution Readiness

SkillMatch AI is being built by a small, focused team of RUET students with complementary strengths across artificial intelligence, full-stack development, and product design. The team combines academic depth with hands-on project experience in machine learning, mobile engineering, and cloud deployment — exactly the capabilities required to move the proposed MVP from concept to a working pilot within the twelve-month roadmap outlined above.

- **AI & Data Lead.** Owns the skill-gap model, LLM prompting pipeline, and vector-based job matching engine.
- **Mobile & Backend Lead.** Builds the Flutter application and the FastAPI/Node.js backend services.
- **Product & UX Lead.** Runs user research at RUET, designs the onboarding flow, and coordinates the pilot with partner departments.
- **Growth & Partnerships Lead.** Engages universities, EdTech partners, and recruiters; drives community acquisition on campus.

The team is supported by mentorship from senior faculty and industry practitioners, and by RUET’s innovation ecosystem — giving the project immediate access to a controlled pilot environment with several thousand eligible student users on day one.

18. Future Scope

- Regional expansion across South Asia — India, Nepal, Sri Lanka, Pakistan.
- Full Bangla language support with local dialect handling.
- Blockchain-verified skill badges recognized by partner employers.
- AR/VR-based immersive interview simulations for advanced roles.
- Integration with government skill-development programs and NSDA.

19. Conclusion

SkillMatch AI is more than an app — it is an equal-access career infrastructure for the next generation of Bangladeshi talent. By fusing personalized AI mentorship with real-time job intelligence, it transforms career guidance from a luxury service into a public utility, available to every student with a smartphone. The idea is technically feasible with today’s AI stack, financially sustainable through a multi-channel revenue model, and socially urgent given Bangladesh’s graduate-unemployment crisis. With institutional backing and pilot deployment at RUET, SkillMatch AI can move from concept to national impact within twelve months.

Ideas today, impact tomorrow. SkillMatch AI turns that promise into a measurable outcome — a smarter, more employable, more confident generation of graduates.